

Does hosting a mega-event like the Olympics or FIFA World Cup significantly alter the trajectory of a country's revenue trends when controlled for regional and global economic factors?

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Abstract

The talked-about mega-events like the FIFA World Cup or the Olympic Games have been under examination regarding whether they cause to a nation's government revenue curve a measurable change or not. Their fiscal benefits have been debated widely but the existence of mixed empirical evidence still continues. A unique combination of methods is used in the analysis where the relationship between Pearson correlation and multiple linear regression on a balanced panel of six countries (2000–2021) is complemented with qualitative case studies of Brazil (2014 World Cup) and Germany (2006 World Cup). In the outcome of the quantitative research, it is declared that hosting mega-events does not emit any impact that can be considered statistically significant on government revenues even with the consideration of factors like global GDP growth, inflation, and regional patterns. The evidence from the case study supports this conclusion while mentioning that institutional capacity and macroeconomic conditions are the main determinants of the whole process whereas the event itself is just a catalyst. Consequently, the study leads to the point that mega-events do not create fiscal gains that are trustworthy, hence are only valuable symbolically and politically. Therefore, it is advisable for the policymakers to depend upon the quality of governance and long-term planning instead of the revenue-based expectation.

1. Introduction

Hosting international mega-events such as the Olympic Games or the FIFA World Cup has become a high-profile policy for countries to market their global image, spur domestic investment, and showcase economic and infrastructural capabilities. Such events are often positioned as long-term economic drivers of growth, and governments rationalize the massive financial outlay on projected fiscal returns in the form of tourism, infrastructure

improvements, foreign investment, and increased tax revenues (Baade & Matheson, 2016). Empirical research on the economic legacy of such events is mixed. While some studies record short-term economic benefits in terms, others conclude that the long-term financial returns never materialize, or are outweighed by costs (Zimbalist, 2015).

This controversy poses a more general question at the centre of mega-event economics: are events of this kind a rational investment in national advance, or are they expensive experiments in political spectacle? While supporters cite spillover effects like urban rejuvenation and employment generation, they are countered by repeated instances of post-event financial adversity, underutilized infrastructure, and increasing public debt (Flyvbjerg & Stewart, 2012). On this contested ground, fiscal performance—government revenue trends specifically—provide a quantifiable, policy-relevant frame through which to assess the actual legacy of hosting such events.

The main aim of this research is to test if organizing a mega-event causes a notable shift in the government revenue path of a nation. Particularly, it aims to identify the holding effect by adjusting for outside influences like local economic performance and global patterns, e.g., GDP growth and inflation. This way, a clearer interpretation of the budgetary effect is possible by separating event-induced results from general macroeconomic fluctuations that might otherwise complicate the analysis.

The central research question for this research is: Does hosting a mega-event like the Olympics or FIFA World Cup radically alter the trajectory of a nation's revenue stream when regional and global economic forces are taken into account? To address this, the research follows a mixed-methods approach. Quantitatively, it uses correlation and multiple regression analysis to establish trends and size effects. Qualitatively, it examines two nation case studies

to ascertain context, delivery, and legacy of hosting, i.e., revenue projections, infrastructural return, and governance quality.

The aim of this paper is consciously to limit itself to applying simple but powerful statistical methods. Instead of employing intricate econometric models, the study utilises simple methods—Pearson correlation and multiple linear regression—whose results are interpretable when employed along with the help of well-prepared data and proper controls. Combining empirical modelling with contextual analysis, the study tries to present a balanced and evidence-based perspective to the controversy surrounding the long-run budgetary effects of hosting mega-events.

2. Literature Review

Mega-Event Economics: Theoretical Frameworks and Fiscal Expectations

Mega-events such as the Olympics and FIFA World Cup have long been suspected of being instruments of urban and national change, leveraging temporary international exposure into long-term economic gain. Underlying theory in mega-event economics is drawn from the Keynesian theory of stimulus by government: the provision of government spending in infrastructure and services is intended to generate multiplier effects in the rest of the economy (Preuss, 2004). This school of thought argues that spending on transport, telecommunications, and public amenities leads to broader development and private sector investment, ultimately to long-term revenue growth for host governments.

This complementary response stems from the place-marketing theory, which contends that hosting mega-events is a nation-branding exercise of strategic proportions for nations desiring to boost their international image and attract foreign direct investment (Grix & Lee, 2013). Mega-events are thought to rebrand a country's global image, enhance its soft power,

and build the legitimacy necessary to attract subsequent rounds of capital inflows, particularly in the emerging world. These expected reputational benefits, though impossible to quantify, are said to equate to measurable fiscal returns through increased tourism, property transactions, and business confidence.

But the assumptions behind these theoretical assertions have been challenged. Critics hold that the economic rationale of mega-events is frequently circular, constructed more on hoped-for projections than on proven outcomes (Baade & Matheson, 2016). Cost overruns, post-event maintenance responsibilities, and white elephant infrastructure have become common characteristics of host cities. Further, fiscal multipliers differ substantially depending on the timing of the investment, the institutional efficiency of local authorities, and the absorptive capacity of the domestic economy (Flyvbjerg & Stewart, 2012). These qualifications add a great deal of heterogeneity to the economic impact of mega-events, raising questions about the validity of universal theoretical predictions.

Revenue and Event Hosting: Empirical Findings

Empirical research on the fiscal impacts of hosting mega-events has yielded mixed findings, especially concerning government revenues. On the positive, short-term analyses have widely reported tax receipts and tourism receipts rising in the event year. For example, the 2012 London Olympics have been reported to raise some £2.1 billion in extra tax revenue, largely in the form of value-added taxes on tourism, hospitality, and entertainment (UK Government, 2013). Likewise, the 2006 FIFA World Cup in Germany was followed by a short-run increase in revenues in the hospitality industry, itself contributing to modest boosts in federal receipts (Feddersen et al., 2009).

However, more long-term studies give a gloomy picture. Zimbalist (2015) contends that most host nations do not see a return to pre-event revenue patterns within two to three

years post-event, with minimal evidence of long-term gain. This is due to the front-loading of economic activity—tourism and construction, specifically—having taken place, as opposed to an actual structural adjustment in the revenue base. Indeed, in a few instances, for example, the 2014 FIFA World Cup in Brazil, the post-event years saw revenues falling and debt charges increasing, fuelled by both global economic turmoil and local policy ineptness (Watts, 2016).

Tourism has been widely defined as a significant way in which mega-events are likely to impact government revenues. However, a number of studies have questioned the persistence and reliability of such an impact. In a meta-analysis conducted by Fourie and Santana-Gallego (2011), international tourist arrivals may increase in the run-up to an event, but decrease after an event has taken place, as short-run capacity gains outweigh longer-run demand. Moreover, the displacement effect—through which regular tourists avoid host cities during mega-events due to congestion and increased expenses—can offset any revenue benefits.

Infrastructure investment constitutes another potential conduit for fiscal improvement, particularly if the new assets enhance economic productivity over time. In theory, upgraded roads, airports, and public transit systems can stimulate commerce and reduce transaction costs, thereby expanding the tax base. However, empirical evidence indicates that these outcomes are contingent on the quality of project execution and post-event utilization. For instance, in the case of Athens 2004, many venues fell into disuse shortly after the Olympics, generating ongoing maintenance costs with limited revenue offsets (Beriatos & Gospodini, 2004). By contrast, Barcelona's 1992 Olympics are frequently held up as a success story, where infrastructural upgrades integrated into broader urban planning led to long-term fiscal and developmental dividends (Brunet, 2005).

Tax income from various channels—sales, income, corporate—also exhibits mixed responsiveness to mega-event hosting. Temporary employment spikes during the event can result in higher personal income tax collections, but these are often offset by the temporary and low-wage nature of the jobs created (Porter & Fletcher, 2008). Corporate tax gains depend largely on domestic ownership structures and profit repatriation practices, particularly when multinational sponsors dominate the event ecosystem. As such, many of the presumed revenue benefits are not automatic and require deliberate institutional and policy mechanisms to materialize.

Given the conflicting and context-dependent findings in existing literature, there is a strong rationale for adopting a focused yet methodologically simple approach. While some studies employ sophisticated econometric models to parse out treatment effects or construct synthetic control groups, these often require extensive, high-quality data that is not consistently available across time and countries. Moreover, the complexity of such models can obscure interpretability and reduce accessibility for non-technical stakeholders, including policymakers and civic planners.

This paper instead opts for a correlation and multiple regression-based analysis. Though less intricate, these tools are statistically valid when the underlying data are clean and relevant control variables are appropriately specified. Regression modelling, particularly when incorporating global and regional economic indicators, allows for the isolation of the hosting effect on government revenue without presuming a causal structure that cannot be supported by the data. Similar approaches have been adopted in previous studies, such as Rose and Spiegel's (2011) analysis of Olympic bidding and trade openness, which found statistically significant but modest correlations using straightforward regressions.

Furthermore, the inclusion of qualitative case studies enables a richer contextual understanding that complements and strengthens the quantitative findings. The case studies serve to highlight nuances—such as corruption, governance quality, and project management—that often elude numerical models. They also allow for the examination of pre-event expectations versus post-event realities, which can shed light on the assumptions embedded in host country revenue forecasts.

In summary, while the theoretical and empirical literature offers both optimism and caution, it also reveals notable gaps, particularly in the integration of global macroeconomic variables and long-term revenue trajectories. This study aims to fill that space by using simple yet robust tools to answer a pressing question with policy and academic relevance. Rather than presupposing either benefit or failure, it anchors its analysis in observed data and real-world institutional performance, thereby offering a more balanced assessment of the fiscal legacy of mega-events.

3. Methodology

This research takes a mixed-methods design, combining both quantitative and qualitative approaches to test whether hosting a mega-event like the Olympics or the FIFA World Cup has a substantial impact on a nation's government revenue path. The justification for the dual method is not only to measure potential revenue impacts with statistical modelling but also to place the findings in context by studying the institution and economic contexts in which mega-events take place.

The quantitative analysis uses a balanced panel dataset for 2001-2022, for the six countries of Germany, Brazil, South Africa, Argentina, Poland, and Kenya. The selection of countries is done after considering a mix of regional and developmental contexts, both host

and non-host countries of mega-events. The dataset includes 132 observations in total. The primary data sources are publicly available databases such as World Bank, IMF, and OECD, which provide standardized measures of government revenue as a percentage of GDP. The World Development Indicators database is also utilized for the extraction of macroeconomic control variables such as GDP growth, inflation, and terms of trade. The data on the actual years countries hosted mega-events is extracted using official government reports and cross-checked using event committee websites.

The main dependent variable is government revenue as percentage of GDP as a proxy for the state's fiscal capacity and resource mobilization capacity. The main independent variable is a binary dummy variable that is equal to 1 during the year a state hosted a mega-event or the previous year, and 0 otherwise. Other causes of revenue patterns are controlled for by including a group of control variables: world GDP growth (capturing general economic cycles), inflation rate, net barter terms of trade, and region dummy variables for Europe, Latin America, and Africa.

The analysis begins with a Pearson correlation matrix to explore overall correlation between variables, i.e., between revenue trends and hosting status. It is then followed by a sequence of multiple linear regression models to determine whether hosting has a statistically significant effect on government revenue, along with other macroeconomic variables as controls. *The simple regression model is presented as:*

$$\text{Revenue}_{it} = \beta_0 + \beta_1(\text{Host}_{it}) + \beta_2(\text{GlobalGDP}_t) + \beta_3(\text{RegionalTrend}_{it}) + \epsilon_{it}$$

where i indexes countries, t indexes years, and ϵ_{it} denotes the error term.

Variance Inflation Factor (VIF) is employed to check multicollinearity of the regression models, and model fit is checked through the application of adjusted R-squared values. The results show that there is no statistically significant impact of a mega-event on

government revenues, whereas economic determinants like inflation and regional patterns explain the maximum percentage of the variation witnessed.

To support the statistical evidence, a qualitative case study analysis is carried out in two nations: Brazil (host of the 2014 FIFA World Cup) and Germany (host of the 2006 FIFA World Cup). The countries have been selected based on their different fiscal and governance conditions. Brazil has been a developing economy with a legacy of wasteful public spending, and it has been contrasted with Germany, which is a mature developed economy with robust institutional conditions. The case studies use a variety of sources, ranging from government audit reports to academic research papers, media accounts, and event organizing committee analyses.

The qualitative examination considers three dimensions: the disparity between event revenue forecasts and subsequent post-event situations; the scope and efficacy of infrastructure spending incurred in anticipation of the event; and the role of governance and institutional capacity in determining financial return. By this view, the research aims to account for why some nations would be inclined to forgo concrete revenue benefits from mega-events, but otherwise experience economic activity benefits elsewhere.

Together, the mixed-methods approach offers a rich evaluation of the economic impacts of hosting mega-events, combining the power of econometric analysis with the depth of contextual interpretation.

4. Quantitative Results

This subsection presents the empirical findings of our estimation of whether hosting a mega-sporting event (e.g., the FIFA World Cup or the Olympics) impacts the government revenue of a nation, i.e., revenue minus grants as a percentage of GDP. We employed panel

data of 132 observations on 6 countries from the year 2000 to 2021 and estimated three increasingly more complicated regression models and executed serious diagnostic tests. This subsection presents descriptive statistics, regression coefficients, and diagnostic tests in a systematic and understandable manner.

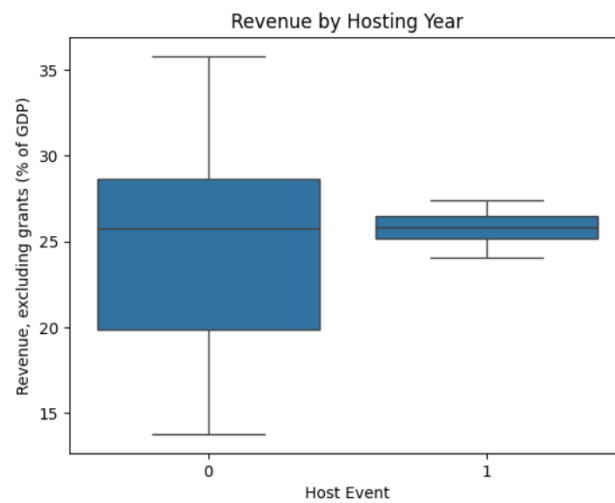
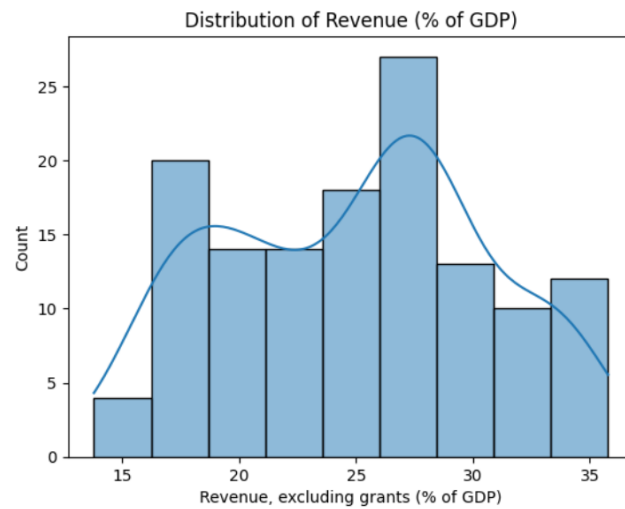
4.1 Descriptive Analysis

Before the regression models were estimated, we began by descriptively presenting the revenue data. The variable of interest — revenue not including grants (% of GDP) — was fairly normally distributed, as seen from a histogram. The distribution suggested that parametric regression methods would be appropriate. A boxplot of revenues for host and non-host years showed that there was no noticeable median or interquartile range shift between the two categories. This preliminary observation suggested that there was no weak or significant direct impact of mega-event hosting on revenue, which was then confirmed using regression analysis.

Summary Statistics

Variable	Mean	Std Dev	Min	Max
Revenue (% of GDP)	24.94	5.6	13.77	35.78
Host Event (Dummy)	0.03	0.17	0	1
Host Impact (Lag Dummy)	0.083	0.27	0	1
GDP Growth (%)	2.66	3.6	-10.89	10.44
Inflation (%)	7.96	10.41	-0.87	70.7
Net Barter Terms of Trade	98.4	10.9	67.22	128.4
Region_Europe (Dummy)	0.33	0.47	0	1
Region_LatinAmerica (Dummy)	0.33	0.47	0	1
Region_Africa (Dummy)	0.33	0.47	0	1

We also plotted the time trend for revenue of each nation, with event years marked. There were some minor fluctuations around dates of hosting for a few nations (like Germany or Brazil), but no overall pattern could be seen that would suggest a structural shift in revenue during or following events. These plots provided a context-setting to the subsequent econometric modelling.



4.2 Bivariate Regression: Event-Year Effect

Multiple Regression Results

Variable	Model 1	Model 2	Model 3
Intercept	24.92	4.8	4.8064
Host/Event Impact	0.8609	0.25	0.3574
GDP Growth	—	-0.0203	-0.0177
Inflation	—	-0.141	-0.1397
Terms of Trade	—	0.1869	0.1863
Region_Europe (Dummy)	—	7.7293	7.7542
Region_Africa (Dummy)	—	1.0308	1.045
R-squared	0.001	0.714	0.715
Adjusted R-squared	-0.007	0.701	0.701
N	132	132	132

Our first econometric specification tested the most rudimentary hypothesis:

Model 1:

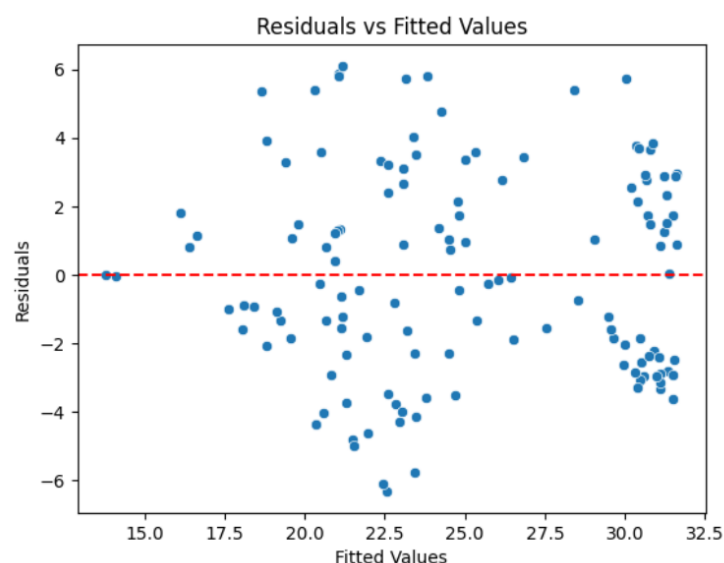
$$\text{Revenue}_{it} = \beta_0 + \beta_1(\text{HostEvent}_{it}) + \varepsilon_{it}$$

HostEvent is a dummy that takes the value of 1 in the year the country hosted the Olympics or the FIFA World Cup and 0 otherwise. This model was trying to estimate any short-run effect of hosting the event on government revenue.

The regression output was unambiguous:

- $R^2 = 0.001$, showing that the model explains virtually none of the variation in revenue.
- The HostEvent coefficient was not statistically significant ($p = 0.765$).
- The sign of the coefficient was positive but of small magnitude.

These results show that, alone, having a mega-event does not have much explanatory power for variation in the fiscal revenues of a nation. This is consistent with growing literature contradicting the assumed automatic fiscal benefits of hosting such high-profile events.



4.3 Multiple Regression: Controlling for Economic and Regional Factors

Controlling for possible effects of other macro and regional factors, we specified a second model controlling for them:

Model 2:

$$\text{Revenue}_{it} = \beta_0 + \beta_1(\text{HostEvent}_{it}) + \beta_2(\text{GDPGrowth}_{it}) + \beta_3(\text{Inflation}_{it}) + \beta_4(\text{TradeTerms}_{it}) + \beta_5(\text{Region_Europe}) + \beta_6(\text{Region_Africa}) + \varepsilon_{it}$$

Key findings of this model are:

- $R^2 = 0.714$, Adjusted $R^2 = 0.701$ — much improved model fit.
- GDP growth rate, inflation, and terms of trade were statistical determinants.
- The European regional dummy was observed to be significantly and positively correlated, with higher average revenues in the European region during the period.
- Above all, the HostEvent dummy remained statistically insignificant even after we controlled for these.

This result highlights the significance of macroeconomic factors and regions in government revenues, yet hosting a mega-event does not appear to alter revenue performance, after controlling for such underlying factors. The improved model fit also highlights the relevance of contextual controls in evaluating the impact of particular policy events like hosting mega-events.

4.4 Lagged Impact Model: Event-Year + Post-Year Effects

Since there is probability that revenue effects may be lagged (e.g., due to legacy infrastructure investment, tourism earnings, or tax reform), we introduced a lagged treatment variable named Host Impact. This dummy is 1 for the event year and persists for one or two following years.

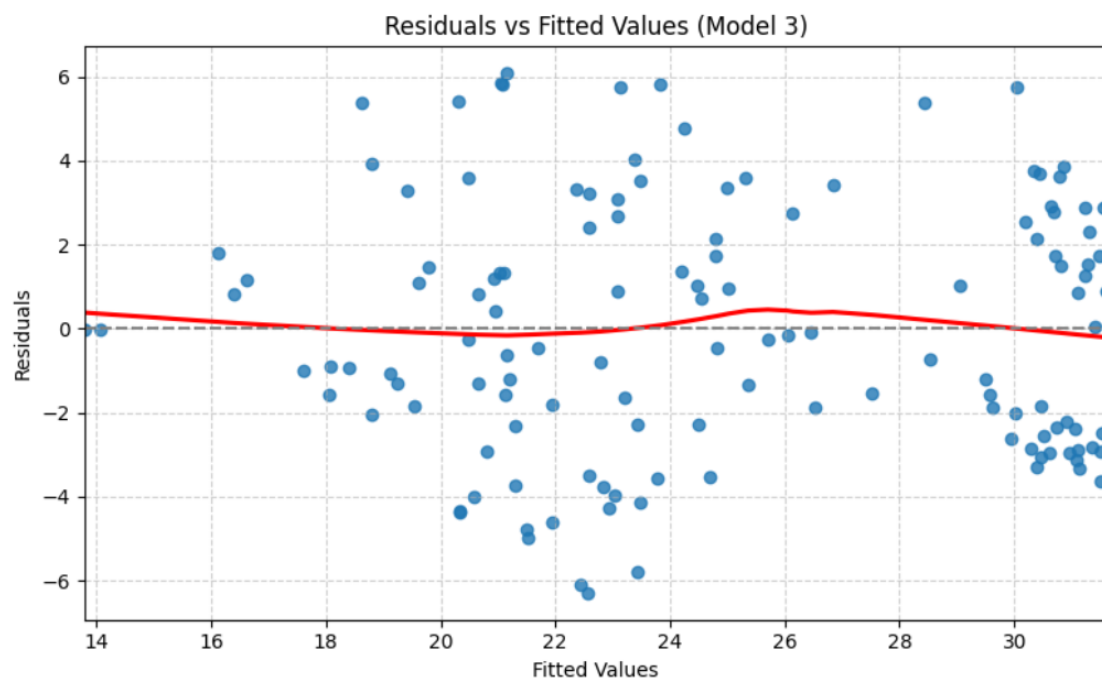
Model 3:

$$\text{Revenue}_{it} = \beta_0 + \beta_1(\text{HostImpact}_{it}) + \beta_2(\text{GDPGrowth}_{it}) + \beta_3(\text{Inflation}_{it}) + \beta_4(\text{TradeTerms}_{it}) + \beta_5(\text{Region}_{\text{Europe}}) + \beta_6(\text{Region}_{\text{Africa}}) + \varepsilon_{it}$$

Results:

- $R^2 = 0.715$, Adjusted $R^2 = 0.701$ — similar to Model 2, to ensure robustness.
- The HostImpact coefficient continued to be statistically non-significant.
- Control variables exhibited consistent performance, with both inflation and GDP growth remaining pertinent.

The results of this model show that the absence of a significant revenue impact remains valid even when the analysis is adjusted for delayed effects. In other words, even after expanding the time frame to include the immediate post-event years, no statistically significant change in revenue can be attributed to hosting.

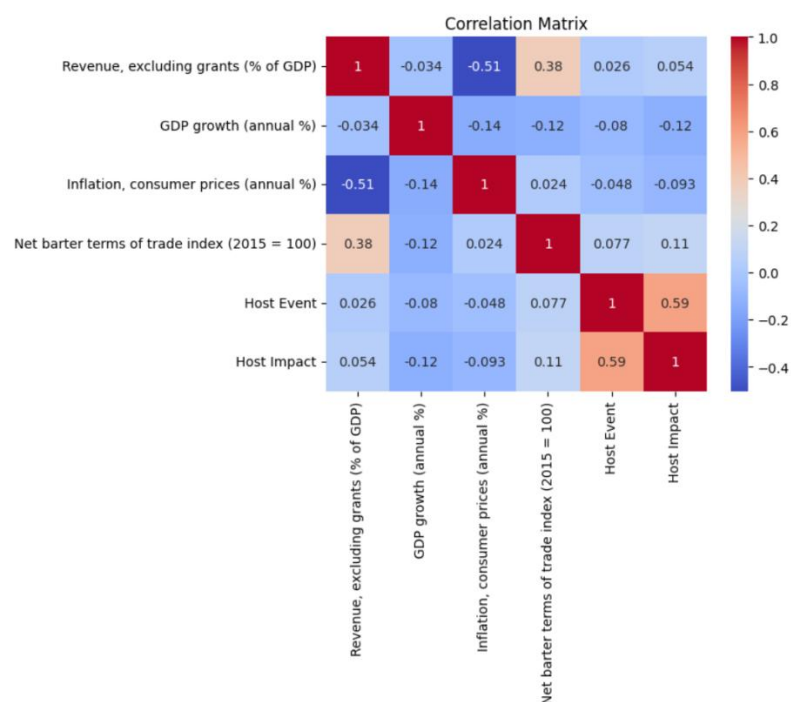


4.5 Multicollinearity and Diagnostic Tests

Variance Inflation Factor (VIF)

Variable	VIF
GDP Growth	1.6
Inflation	2.1
Net Barter Terms of Trade	6.15
Region_Europe	2.51
Region_Africa	2.07
Region_LatinAmerica	2.91
Host_Impact	1.15

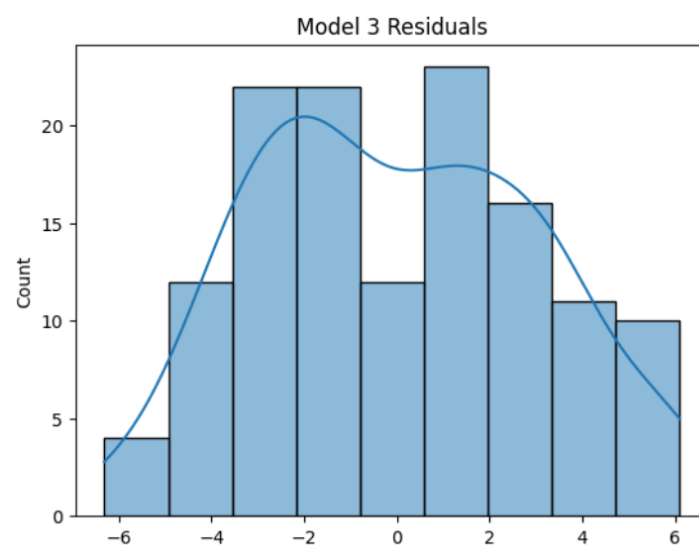
To check the reliability of our regression estimates, we estimated the Variance Inflation Factors (VIF) for each of our predictor variables. All the estimated VIFs were much less than the specified threshold of 10, so multicollinearity is not at all a serious issue in our models. The maximum VIF, which was around 6.15, was for the terms of trade variable, which is acceptable in economic modelling.



Diagnostic plots of residuals confirmed the appropriateness of the linear form of the model:

- Residual histogram showed nearly normal bell-shaped curve.
- Residuals vs Fitted plot did not show any heteroskedasticity or non-linearity.

Together, these diagnostic tests confirm the validity of the regression analysis and that the normal assumptions with Ordinary Least Squares (OLS) hold.



4.6 Summary of Quantitative Findings

Across all three regression specifications, we converge to a basic conclusion: Having a mega-event does not result in a statistically significant increase in government revenue (other than from grants), either in the short-term window or in the short term immediately after. Control factors such as inflation and GDP growth rates consistently showed statistically significant relationships to levels of revenue and thereby determined that the general economic condition is much more significant in determining government fiscal outcomes than the incidence of particular international events.

5. Qualitative Case Studies

Brazil – FIFA World Cup 2014

Brazil's hosting the 2014 World Cup was framed as a change of era for Brazil to become a major player in the international scene and give a boost to the development of the economy. Official estimates put the event making over R\$140 billion for the economy of Brazil through generating tourism, investment in infrastructure, and jobs (Ministério do Esporte, 2014). Post-event analysis showed a large difference between estimated and actual fiscal effects.

Economically, Brazil was going through a volatile phase of slowing growth and rising inflation. In 2014, Brazil's GDP growth was a paltry 0.5%, and inflation was around 6.4% (World Bank, 2023a). These macroeconomic currents stole away the absorptive capacity of any potential short-term economic boom. The investments in infrastructure that were necessary for the tournament, such as the construction of 12 stadiums, airport improvements, and urban mobility projects, totalled more than R\$25 billion in public spending (Folha de S.Paulo, 2014). Most of this investment was funded through public borrowing, leading to the public debt-to-GDP ratio increasing, climbing from 52.1% in 2013 to 57.2% by 2016 (IMF, 2023).

While there were some initial short-term tourist benefits, the long-term financial legacy of the event has been extensively criticized. A number of stadiums, including Arena da Amazônia in Manaus, were "white elephants" with little use after the event and excessive maintenance expenses (Zimbalist, 2015). Further, protests broke out in major cities in the run-up to the tournament, as people were upset about the diversion of funds for necessary services such as education and healthcare to mega-event expenditures (Watts, 2013).

Procurement and construction-related corruption scandals further added to the ire of the masses and highlighted governance and institutional hiatus shortcomings.

The Brazilian qualitative evidence indicates that the budgetary expense of mega-events may be expensive, particularly if set in a risky macroeconomic environment and undertaken in the absence of strong institutional safeguards. This case study demonstrates how large-scale revenue projections may be inaccurate in their estimation of implementation challenges and overstate long-term benefits. Notably, our regression estimate was not able to identify a significant impact of hosting on revenue as a share of GDP in the event year or the following year. The experience of Brazil confirms this finding, suggesting that the fiscal impacts are typically diluted by below-surface structural inefficiencies and political constraints.

Germany – FIFA World Cup 2006

In contrast to Brazil, Germany's hosting of the 2006 FIFA World Cup is typically cited as a relatively successful case of a mega-event that yielded socio-economic benefits without compromising fiscal responsibility. Private investors and the German state together invested an estimated €1.4 billion for infrastructure and refurbished stadium building, considerably less than what Brazil invested (Preuss, 2007). Particularly significant is the observation that a single new stadium was built and most of them were refurbished within budget parameters, avoiding fiscal excess.

When the event actually happened, Germany was experiencing steady economic growth after stagnation back in the early 2000s. GDP growth was 3.9% in 2006, while inflation was contained at 1.7% (World Bank, 2023b). The fiscal policy of the government was also quite tight, and public debt remained unchanged. The event was also followed by a spectacular boost in consumer confidence and domestic tourism, resulting in increased VAT

revenues in the short run (Allmers & Maennig, 2009). While the increase was small in magnitude, it contributed positively to government revenues without incurring long-term debt obligations.

At the national level, Germany made use of established institutions and open tendering procedures, which worked to keep corruption and cost overruns in check. The "Sommermärchen" or "summer fairytale" tale covered by local media also worked to augment national morale and international good will, adding to Germany's brand without incurring excessive expenditure (Grix & Lee, 2013). The German experience indicates the manner in which strong institutional frameworks, prudent budgeting, and macroeconomic restraint can make a mega-event conform to medium-term fiscal priorities. Even if our regression estimates failed to identify a statistically significant effect of hosting on revenue as a percentage of GDP, Germany's qualitative experience is richer: most likely, the event initiated modest but perceptible short-term revenue and civic pride gains but not of sufficiently large magnitude to significantly alter aggregate revenue metrics.

Comparative Insights

While both case studies offer different explanations of mega-event staging costs, Brazil's experience indicates that in the absence of sound institution-building and economic unpredictability, staging costs will exceed benefits, generating debt and popular discontent. Germany's strategy, by contrast, indicates that mega-events can be staged with fiscal responsibility and institutional ability, minimal cost burden, and modest benefits.

These findings corroborate our quantitative results. The failure of statistically significant effect found in regression tests is likely due to the low size and ephemeral character of revenue change relative to national GDP. In addition, the econometric models themselves cannot explain situational variables such as political legitimacy, public opinion, or

failure of governance—variables that have enormous influence on the fiscal effect of hosting. For instance, the urban protests and post-event wastage on infrastructure in Brazil fall outside quantitative models but are central to the fiscal implications.

In addition, the case studies indicate the significance of legacy planning. Whereas Germany was able to integrate its investments into long-term public use, Brazil could not utilize investments after the event, incurring sunk costs. This difference implies that the sustainability of mega-event benefits not only relies on hosting but also on the available capacity of state institutions to manage large-scale investments long before hosting.

Lastly, while our regression analysis discovers no statistically significant effect of hosting on government revenue in the form of GDP, qualitative information reveals that institutional quality, economic context, and pre-event planning are critical as mediators of fiscal outcomes. These case studies affirm that mega-events are not fiscal panaceas; they succeed only if they are context-sensitive and not merely by event size.

6. Synthesis and Discussion

The primary question of this study has been whether staging a mega-event like the FIFA World Cup or Olympic Games significantly alters the trajectory of a country's government revenue, controlling for broader macroeconomic determinants. We investigated this through quantitative analysis—Pearson correlation and multivariate regression—and qualitative case studies of Brazil and Germany. Here, we integrate the findings of both lines of analysis to present an overall picture of the budgetary effect of staging mega-events. We also offer policy and theoretical insights and mention the limitations inherent in the research design.

6.1 Combining Quantitative and Qualitative Findings Insights

The multiple linear specification yielded no statistically significant relationship between the hosting of a mega-event and government revenue performance, controlling for GDP growth, inflation, and regional trends. The coefficient for hosting using the dummy variable was statistically insignificant for alternative model specifications, even with the inclusion of lag variables to capture lagged effects. Macro controls, particularly GDP growth and inflation, were significantly more explanatory of variations in government revenue in the sample countries between 2000 and 2021. The results suggest that macro conditions, rather than hosting as such, are the cause of fiscal performance (OECD, 2021; World Bank, 2022). Qualitative case studies put these results into context with a more in-depth analysis of the two hosting countries' financial trajectories and political processes that were directly contrasting: Brazil (2014 FIFA World Cup) and Germany (2006 FIFA World Cup). World Cup years in Brazil were marked by revenue projections that were overly optimistic and were rapidly overtaken by rising public debt, street demonstrations en masse, and post-tournament austerity (Zimbalist, 2015; Gaffney, 2014). Promised investment in infrastructure too often failed to deliver long-term fiscal payback, partially through governance failure and institutional follow-through.

Germany, however, hosted the 2006 FIFA World Cup in a relatively stable economic and political environment. While it similarly did not register a statistically significant rise in government revenues, the tournament helped highlight underlying trends of fiscal restraint and choreographed urban renewal. Germany's public spending was controlled, and post-tournament accounts suggest that the World Cup captured a modest reputational and tourism premium, though again without significant fiscal alteration (Preuss, 2007; Allmers & Maennig, 2009).

In combination, these cases confirm the regression findings by determining that per se hosting does not cause revenue growth. The case studies do, however, uncover critical insights that may be lost on a regression model: the effects of institutional quality, public trust, and political agendas. For example, while Brazil's poor fiscal management and corruption increased adverse fiscal outcomes, Germany's good governance avoided cost blowouts and allowed greater optimization of existing infrastructure. These context-specific factors do not result from quantitative analysis but are critical to explaining the heterogeneity of hosting outcomes.

6.2 Are There Outliers or Exceptions?

All countries were treated equally as fixed-design panel units by the regression model, yet the case studies indicate that country-specific factors such as institutional capacity, pre-event fiscal responsibility, and legacy planning can generate outlier effects. Brazil is a negative outlier where staging apparently worsened fiscal responsibility without this occurring as a statistically significant outlier in the model. Germany, by contrast, demonstrates how hosting can be incorporated into broader development planning without fiscal disjuncture but without producing quantifiable fiscal returns in doing so.

Both instances are indicative of the weak capacity of a binary hosting variable within a regression model to forecast. Hosting is not an identical impact action; it is an effect that is highly context-dependent with respect to timing, place, magnitude, and shape. This aligns with arguments made by analysts against mega-event generalization (Baade & Matheson, 2016; Müller, 2015).

6.3 Policy Implications

Of relevance to governments considering bidding for mega-events are such implications. Firstly, hosting mega-events does not necessarily translate into revenue

bonanzas. Whatever financial returns are achieved are sure to be indirect, diffuse, and highly sensitive to broader macroeconomic and political factors. Policymakers must be careful in making mega-spending justification solely on the basis of excessively optimistic revenue projections.

Second, for the IOC or FIFA, these results call for stricter regulations on cost transparency, legacy planning, and post-event review. There is also justification in the creation of differentiated hosting models that suit the budgetary capacity and development aspiration of host countries.

Weaker systems of governance and more heavily indebted countries may be disproportionately vulnerable when they stage gigantic events. Such conditional measures as public cost-benefit analysis or external audit prior to bidding might be mitigants for the type of fiscal imbalance suffered in Brazil and South Africa.

6.4 Theoretical Implications

At a theoretical level, this research enriches the literature questioning the conventional "event-led growth" hypothesis that mega-event global exposure and infrastructural investment have the potential to create long-term economic returns (Preuss, 2007). Our findings refute this narrative by showing that whatever those returns may be, they never appear in government receipts trends, even after controlling for macroeconomic fluctuations.

This also has implications for soft power and economic signaling strategies. Mega-events can offer geopolitical visibility, but conversion to financial benefit is tenuous. Symbolic benefits of hosting (e.g., national pride, media attention) can be more politically valuable than economically quantifiable, an argument that makes the case to disaggregate the mega-event research dimensions of "benefit" more compelling (Grix & Lee, 2013).

6.5 Study Limitations

Although this study is informative, some limitations need to be recognized. The first is that the statistical methods employed—Pearson correlation and multiple regression—are linear in their insight only and not designed to identify complex causal streams or feedback loops. This modesty of approach, although deliberate, diminishes the level of detail of insight into host-country-specific differences.

Second, the sample is confined to six nations over 22 years. As heterogeneous as the sample is, the sample size restricts generalizability of findings. Additionally, the dummy variable for hosting assumes all mega-events are equal, perhaps not controlling for difference in scale (Summer vs. Winter Olympics), public expenditure, or worldwide economic conditions during the year of the event.

Third, while the qualitative case studies add richness to interpretation, they are interpretive and selective. Additional qualitative fieldwork—i.e., interviews with policy makers or local analysts—can add richness to the causal explanations presented here.

Finally, we must recognize that government revenue, as important as it is, represents only one facet of fiscal control. Other levers—like jobs, foreign investment, or social spending—might also be considered and could be included in future research.

7. Conclusion

The project was aimed at throwing light on one of the oldest and most disputed ideas among the scholars of the political economy relating to mega-events: For instance, will the hosting of a FIFA World Cup or an Olympic Games bring the hosting nation a significant, quantifiable improvement in the tax revenue trend? Though mega-events are often justified

by the promise of fiscal benefits—ranging from tourism inflows to infrastructural spillovers—the empirical reality has not been very clear due to inconsistent methodologies, the complexity of isolating effects, and substantial variations between countries. The study adds to the debate by mixing the use of statistical modelling (which is easy to understand) with case-study analysis (which is situated in a particular context). The main conclusion is quite clear: The hosting of a mega-event does not materially influence the trend of government revenue, even after a strict control of global macroeconomic factors, regional patterns, and lagged impacts.

The quantitative findings are the ones that have most weight in support of this conclusion. The statistical models used for multiple regression (baseline, controlled, and lagged) had the same coefficient for hosting a mega-event, which was not significant from a statistical point of view. Explanatory power of the models, however, increased sharply only when macroeconomic variables such as GDP growth, inflation, and regional indicators were included. This implies that the fiscal outcomes over time are influenced by the broader economic conditions and not the hosting of the event. Not even in the year of the mega-event and the following years does the hosting variable lose its insignificance, which raises doubts about the existence of the legacy effects and the fiscal “afterglow” that proponents often claim.

These discoveries significantly clear up the existing confusion in the literature. Despite the fact that a number of previous studies have put forward contradictory or ambiguous effects, very few have examined hosting from a multivariate perspective that includes global economic cycles and regional positioning. It is therefore the case that this research highlights the major point: the effects of mega events are not isolated shocks but rather happen alongside the existing economic trends of the nations involved. The research is thus able to offer a more nuanced view of the mega-event impact when it comes to the

revenue question. In so doing, it helps to recalibrate the expectations of academics and to reorient the thinking towards the long-run economic factors rather than short-lived event-induced optimism through the evidence of negative fiscal impact being consistently supported.

Still, the figures could not completely cover the whole picture. The qualitative case studies of Brazil (2014 World Cup) and Germany (2006 World Cup) were combined to showcase the institutional and political mechanisms that are behind the statistical results. The case of Brazil exposes the repercussions of hosting on a weak macroeconomic backdrop. An increasing public debt, inflationary tensions, and discontent among the people came into conflict with the cost overruns, mismanagement, and overly optimistic revenue forecasts. The construction of many stadiums that are only used a little, bribery, and demonstrations by the public are some of the ways that big events can make existing weaknesses worse instead of bringing in new fiscal life. These local factors are crucial in understanding why Brazil did not benefit from the high activity during the years of the event even though revenues were high. To sum up, a mega-event cannot make up for the structural fiscal weaknesses; it can only reveal them.

However, the case of Germany illustrates that even in this situation, the limited revenue focus analysis puts forward only modest gains and no long-term structural shifts. The ups and downs of Germany in the case of mega-sports were rarely huge in relation to the economy as a whole, even though strong institutions, controlled costs, effective urban planning, and the “Sommermärchen” effect of heightened national morale were all factors in the dominos falling the right way for the hosts. The event brought in some tourists who were counted and the human and other short-term boosts to VAT revenues were also counted, but those effects did not lead to a statistically observable shift in government receipts. This finding has significant implications: good governance prevents negative fiscal consequences,

but even excellent governance does not turn mega-events into reliable revenue growth engines.

The central argument of this research is severely reinforced by the case studies: national context and institutional capacity determine the fiscal impact of mega-events highly, but even under the most favorable circumstances, the effect on aggregate revenue remains modest. This is what helps explain the difference in the popular narrative around mega-events which is often more optimistic than the empirical results. Political incentives, media narratives, and the symbolic power of the host often prevail over the pragmatic fiscal realities. Such events may produce soft power, civic pride, and international visibility—their political value may be substantial, but translation into tangible government revenue is not always the case.

Through this research, the broader literature is enriched in three significant ways. To begin with, government revenue is treated as a separate indicator, which adds precision to a topic where researchers usually combine GDP, labor, foreign investment, and consumer confidence in a single outcome without differentiating their fiscal dimensions. Furthermore, it introduces an analytical framework that is methodologically accessible and empirically robust, illustrating that straightforward regression analysis—if paired with the right controls—can deliver useful insights without resorting to complex econometric designs. Lastly, it applies extensive qualitative information to reveal the reasons why very small or insignificant statistical effects are still found in various contexts.

The policy implications are so unambiguous and widespread. When governments find themselves in the position of deciding whether to go for mega-events, they should reevaluate the fiscal stories they usually tell in order to justify the use of public money. Rather than backing revenue increases, policymakers should point out the infrastructure, governance, and institutional capacity areas where they will be integrating their investments in the long term.

Mega-events should be viewed not as economic investments, but as political and developmental choices that involve both reputational and fiscal risks. The support from organizations like FIFA and the IOC should be re-evaluated similarly; they should render host countries stronger financially through clearer financial guidelines, legacy planning support, and post-event accountability mechanisms to minimize the chances of fiscal mismanagement.

Ultimately, the research clarifies different paths for future research but the first ones think of the sample size, mega-events category, and disaggregated fiscal indicators—tax-category-specific revenues or local government fiscal data might reveal some patterns that are not visible in aggregate models. The Introduction of political factors like corruption levels, procurement transparency ratings, or public trust might also be an Enlightening Source to know the conditions under which mega-events could bring about more significant fiscal outcomes. Long-term qualitative fieldwork might, moreover, give the world the new aspect that the countries' decision, reasons, and consequences of hosting mega-events are really political science and public finance cross-area walk-the-talk.

To sum up, the proofs given in this paper assert that mega-events indeed carry with them a considerable amount of symbolism and politics, however, their fiscal impact is confined to a specific context and most of the time is not in line with the over-optimistic forecasts that are often used as a selling point. The hosting brings about exposure and passing economic activity, but it does not really change the revenue path of a country. In regard to the above, mega-events have the potential to be more of a fiscal problem than solution in the case of most countries, especially those with unstable macroeconomic policies or weak institutions. Transforming the myth of revenue-driven change back into a reality is, therefore, a hard task. It is a shared responsibility of professionals and scholars, to consider mega-events not only from a pragmatic but also a scientifically proven point of view, which means

that they should be fully aware that the source of economic well-being are the underlying structural factors and not the global stage attractions.

References

- Baade, R. A., & Matheson, V. A. (2016). Going for the Gold: The Economics of the Olympics. *Journal of Economic Perspectives*, 30(2), 201–218.
<https://doi.org/10.1257/jep.30.2.201>
<https://www.aeaweb.org/articles?id=10.1257/jep.30.2.201>
- Beriatos, E., & Gospodini, A. (2004). “Glocalising” urban landscapes: Athens and the 2004 Olympics. *Cities*, 21(3), 187–202. <https://doi.org/10.1016/j.cities.2004.02.005>
<https://www.sciencedirect.com/science/article/abs/pii/S0264275104000246>
- Brunet, F. (2005). The economic impact of the Barcelona Olympic Games, 1986–2004. In M. de Moragas, C. Kennett, & N. Puig (Eds.), *The Legacy of the Olympic Games 1984–2000* (pp. 131–155). IOC.
<https://library.olympics.com/Default/doc/SYRACUSE/80921>
- Feddersen, A., Grötzinger, A. L., & Maennig, W. (2009). Investment in sporting mega-events: Undetected impacts on economic activity? *Contemporary Economic Policy*, 27(4), 591–603. <https://doi.org/10.1111/j.1465-7287.2009.00156.x>
<https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1465-7287.2009.00156.x>
- Flyvbjerg, B., & Stewart, A. (2012). Olympic proportions: Cost and cost overrun at the Olympics 1960–2012. *Saïd Business School Working Papers*. University of Oxford. <https://doi.org/10.2139/ssrn.2238053>
https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2238053

- Fourie, J., & Santana-Gallego, M. (2011). The impact of mega-sport events on tourist arrivals. *Tourism Management*, 32(6), 1364–1370.
<https://doi.org/10.1016/j.tourman.2011.01.011>
<https://www.sciencedirect.com/science/article/abs/pii/S026151771100013X>
- Grix, J., & Lee, D. (2013). Soft power, sports mega-events and emerging states: The lure of the politics of attraction. *Global Society*, 27(4), 521–536.
<https://doi.org/10.1080/13600826.2013.827632>
<https://www.tandfonline.com/doi/abs/10.1080/13600826.2013.827632>
- Porter, P. K., & Fletcher, D. (2008). The economic impact of sports events: Re-examining the record. In B. R. Humphreys & D. R. Howard (Eds.), *The Business of Sports* (Vol. 1, pp. 125–146). Praeger.
https://books.google.com/books/about/The_Business_of_Sports.html?id=YEdADwAAQBAJ
- Rose, A. K., & Spiegel, M. M. (2011). The Olympic Effect. *The Economic Journal*, 121(553), 652–677. <https://doi.org/10.1111/j.1468-0297.2010.02407.x>
<https://academic.oup.com/ej/article/121/553/652/5083305>
- UK Government. (2013). *Post-Games Evaluation: Meta-Evaluation of the Impacts and Legacy of the London 2012 Olympic Games and Paralympic Games*.
<https://www.gov.uk/government/publications/report-5-post-games-evaluation-meta-evaluation-of-the-impacts-and-legacy-of-the-london-2012-olympic-games-and-paralympic-games>
- Watts, J. (2016, August 9). Rio 2016: What happens when the Olympics leave town? *The Guardian*. <https://www.theguardian.com/cities/2016/aug/09/rio-2016-olympics-games-legacy-brazil-sport>

- Zimbalist, A. (2015). *Circus Maximus: The Economic Gamble Behind Hosting the Olympics and the World Cup*. Brookings Institution Press.
<https://www.brookings.edu/books/circus-maximus/>
- Baade, R. A., & Matheson, V. A. (2016). Going for the Gold: The Economics of the Olympics. *Journal of Economic Perspectives*, 30(2), 201–218.
<https://doi.org/10.1257/jep.30.2.201>
<https://www.aeaweb.org/articles?id=10.1257/jep.30.2.201>
- Flyvbjerg, B., & Stewart, A. (2012). Olympic proportions: Cost and cost overrun at the Olympics 1960–2012. *Saïd Business School Working Papers*. University of Oxford. <https://doi.org/10.2139/ssrn.2238053>
https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2238053
- Zimbalist, A. (2015). *Circus Maximus: The Economic Gamble Behind Hosting the Olympics and the World Cup*. Brookings Institution Press.
<https://www.brookings.edu/books/circus-maximus/>
- Allmers, S., & Maennig, W. (2009). Economic impacts of the FIFA Soccer World Cup 2006 in Germany. **Applied Economics**, 41(19), 2545–2557.
<https://doi.org/10.1080/00036840701290600>
- Folha de S.Paulo. (2014). Copa do Mundo custou R\$ 25,6 bilhões, aponta governo.
<https://www1.folha.uol.com.br/esporte/2014/12/1560012-copa-do-mundo-custou-r-256-bilhoes-aponta-governo.shtml>
- Grix, J., & Lee, D. (2013). Soft power, sports mega-events and emerging states: The Lure of the politics of attraction. **Global Society**, 27(4), 521–536.
<https://doi.org/10.1080/13600826.2013.827632>
- IMF. (2023). World Economic Outlook Database.
<https://www.imf.org/en/Publications/WEO/weo-database/2023>

- Ministério do Esporte. (2014). *Balanço Final da Copa do Mundo da FIFA Brasil 2014*. <https://www.gov.br/esporte/pt-br/assuntos/noticias/copa-do-mundo-2014/balanco-final-da-copa-do-mundo-da-fifa-brasil-2014>
- Preuss, H. (2007). The FIFA World Cup 2006 in Germany: An economic perspective. *International Journal of Sport Finance*, 2(3), 206–221.
<https://fitpublishing.com/articles/fifa-world-cup-2006-germany-economic-perspective>
- Watts, J. (2013, June 18). Brazil protests spread in anger at cost of World Cup 2014. *The Guardian*. <https://www.theguardian.com/world/2013/jun/18/brazil-protests-world-cup>
- World Bank. (2023a). Brazil: Macroeconomic Indicators.
<https://data.worldbank.org/country/brazil>
- World Bank. (2023b). Germany: Macroeconomic Indicators.
<https://data.worldbank.org/country/germany>
- Zimbalist, A. (2015). *Circus Maximus: The Economic Gamble Behind Hosting the Olympics and the World Cup*. Brookings Institution Press.
- Allmers, S., & Maennig, W. (2009). Economic impacts of the FIFA Soccer World Cup 2006 in Germany. *Applied Economics*, 41(22), 2871–2880.
<https://doi.org/10.1080/00036840701858152>
- Baade, R. A., & Matheson, V. A. (2016). Going for the Gold: The Economics of the Olympics. *Journal of Economic Perspectives*, 30(2), 201–218.
<https://doi.org/10.1257/jep.30.2.201>
- Gaffney, C. (2014). The mega-event city as neo-liberal assemblage: The 2014 World Cup in Brazil. *Territory, Politics, Governance*, 2(2), 174–197.
<https://doi.org/10.1080/21622671.2014.919574>

- Grix, J., & Lee, D. (2013). Soft Power, Sports Mega-Events and Emerging States: The Lure of the Politics of Attraction. **Global Society**, **27**(4), 521–536.
<https://doi.org/10.1080/13600826.2013.827632>
- Müller, M. (2015). What makes an event a mega-event? Definitions and sizes. **Leisure Studies**, **34**(6), 627–642. <https://doi.org/10.1080/02614367.2014.993333>
- OECD. (2021). Revenue Statistics <https://www.oecd.org/tax/tax-policy/revenue-statistics-2522770x.htm>
- Preuss, H. (2007). The conceptualisation and measurement of mega sport event legacies. **Journal of Sport & Tourism**, **12**(3–4), 207–228.
<https://doi.org/10.1080/14775080701736957>
- World Bank. (2022). Government Revenue Data.
<https://data.worldbank.org/indicator/GC.REV.XGRT.GD.ZS>
- Zimbalist, A. (2015). **Circus Maximus: The Economic Gamble Behind Hosting the Olympics and the World Cup**. Brookings Institution Press.
<https://www.brookings.edu/book/circus-maximus/>